MHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU -UGP (Honours) Regular Examination October 2025
Third Semester
Discipline Specific Elective(DSE)
MG3DSEMATA01-Introduction to Data Analysis Using Python
2024 Admission Onwards
Examination Scheme

Duration: 2 Hours **Total Marks :70**

Remember (K), Understand(U), Apply(A), Analyze(An), Evaluate(E),Create(C)

PART A
MCQ
(Answer any 10 questions. Each carries 3 marks. Total: 30 Marks)

Q1. Which keyword is used for handling exceptions in Python? **A/ [CO1]**
Answer: (a.) try

Q2. What is the output of the following NumPy code: `print(a + b)`? **A/ [CO2]**
Answer: (a.) [[6, 8], [10, 12]]

Q3. Which library provides functions for skewness and kurtosis? **U/ [CO3]**
Answer: (c.) scipy.stats

Q4. What is the Chi-square value for observed [12, 8] and expected [10, 10]? **E/ [CO4]**
Answer: (b.) 0.8

Q5. Which function removes the first occurrence of a value in a list? **A/ [CO2,]**
Answer: (b.) remove()

Q6. What is the output if `x = 25` for conditions `if x>50, elif x>20`? **A/ [CO1]**
Answer: (b.) B

Q7. Which of the following imports the math module in Python? **U/ [CO2]**
Answer: (c.) import math

Q8. Which of the following accesses the second element of a list named `numbers`? **A/ [CO1]**
Answer: (a.) numbers[1]

Q9. Default index when not provided is: **U/ [CO4]**

Answer: (a.) Starts at 0

Q10. If `a=10, b=20, c=20`, then the result of `print(c > b and a > b)` is **A/ [CO1]**

Answer: (b.) FALSE

Q11. Which of the following is immutable? **A/ [CO1]**

Answer: (b.) Tuple

Q12. Which computes column-wise standard deviation? **An/ [CO4]**

Answer: (c.) df.std()

Q13. How many points in the following output? **A/ [CO3]**

```
x = [1, 2, 3, 4]
y = [2, 1, 3, 0]
plt.scatter(x, y)
```

Answer: (c.) 4

Q14. Which Pandas method removes rows with NaN values? **An/ [CO4]**

Answer: (b.) df.dropna()

Q15. Pairplot shows ? **U/ [CO3]**

Answer: (a.) Pairwise scatter/hist across variables

PART B
One or Two Sentences
(Answer any 4 questions. Each carries 5 marks. Total: 20 Marks)

Q16. Define definite and indefinite loops. Give one Python example for each. **[A / CO1]**

Answer: A definite loop executes a fixed number of times **1 Mark**, e.g., a `for` loop.
`for i in range(5):`
`print(i)` executes exactly five times - **2 Mark**
An indefinite loop executes until a condition becomes false -**1 Mark**,
e.g., a `while` loop. `while x < 5:  x += 1` continues until `x` reaches 5.- **1 Mark**

Q17. Print the current local time in HH:MM:SS format. **[A / CO2]**

Answer: To display current local time, use the `datetime or time` module:**(1 Mark)**

```
from datetime import datetime
print(datetime.now().strftime("%H:%M:%S"))
```

This prints time in hours, minutes, and seconds format **(4 Marks)**.

or

```
import time

# Get the current local time
current_time = time.strftime("%H:%M:%S")

# Print the time
print("Current Local Time:", current_time)
```

textbf(4 Marks).

Q18. Write the code to compute sample variance of [2,4,4,4,5,5,7,9]. **[An / CO4]**

Answer: Import the `statistics` module **(1 Mark)** and use the `variance()` function **(2 Marks)**:

```
import statistics
data = [2,4,4,4,5,5,7,9]
print(statistics.variance(data))
```

It computes the sample variance (sum of squared deviations divided by n-1) **(2 Mark)**.

Q19. Write the code to compute determinant and inverse of a 2×2 matrix. **[An / CO4]**

Answer: Import `numpy` for linear algebra **(1 Mark)**.

Use NumPy's linalg functions **(1 Marks)**:

```
import numpy as np
a = np.array([[1,2],[3,4]])
print(np.linalg.det(a))
print(np.linalg.inv(a))
```

Useful for solving linear systems and matrix computations **(3 Mark)**.

Q20. Write the code to print all elements of a 2D list row-wise. **[A / CO2]**

Answer: Use nested loops to iterate rows and items **(1 Mark)**:

```
matrix = [[1,2,3],[4,5,6]]
for row in matrix:
  for item in row:  print(item, end=' ')
  print()
```

This prints all elements row by row **(4 Marks)**.

Q21. Using the built-in `tips` dataset, plot a boxplot of tip amounts. **[E / CO3]**

Answer: Load the dataset and call Seaborn's boxplot **(2 Mark)**:

```
import seaborn as sns
tips = sns.load_dataset("tips")
sns.boxplot(x=tips["tip"])
```

A boxplot shows median, quartiles, and outliers of the `"tip"` values **(3 Marks)**.

PART C
Short Essays
(Answer any 2 questions. Each carries 10 marks. Total: 20 Marks)

Q22. Explain operator precedence in Python with suitable examples. Why is it important? **[A / CO1]**

Answer: Operator precedence defines the order in which operations are evaluated **(1 Mark)**.

Giving an examples **(1 Mark)**.

Listing common mathematical operators in Python in accordance to their preferences - **(8 Marks)**.

Q23. Write a code to
(a.) Create an array of integers and find its maximum, minimum. ,
(b.) Also find the average of members in the array
[A / CO2]

Answer: Import NumPy **(1 Mark)** and use its built-in function. **(2 Marks)**:

```
import numpy as np
arr = np.array([25, 30, 20, 15, 10])
print(arr.max(), arr.min(), arr.mean())
```

This prints the maximum, minimum, and average values of the array **(7 Mark)**

Q24. Write a Python code for a grade distribution pie chart.
Data [A=25, B=30, C=20, D=15, E=10] **[An / CO3]**

Answer: Use Matplotlib to draw a pie chart **(2 Mark)**:

```
import matplotlib.pyplot as plt
grades = ["A", "B", "C", "D", "F"]
values = [25, 30, 20, 15, 10]
plt.pie(values, labels=grades, autopct="%1.1f%%")
plt.title("Grade Distribution")
plt.show()
```

Displays the percentage distribution of grades **(8 Marks)**.

Q25. Create a dataset of 5 students with "Name" and "Gender",
then convert "Gender" into one-hot encoded columns. **[E/CO4]**

Answer: Create a DataFrame and apply one-hot encoding **(2 Mark)**:

```
import pandas as pd
data = {"Name":  ["A", "B", "C", "D", "E"], "Gender":  ["M", "F", "M", "F", "M"]}
df = pd.DataFrame(data)
df = pd.get_dummies(df, columns=["Gender"])
```

This converts the "Gender" column into binary columns such as `Gender_M` and `Gender_F` **(8 Marks)**.

Note: **Any other correct method or logically equivalent code should be awarded full credit. Stepwise marks may be assigned by the examiner based on correctness and completeness of each part.**
